LECTURE 8

Ex

Show $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n+3}$ is conditionally convergent.

Consider $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{2n+3} \right| = \sum_{n=1}^{\infty} \frac{1}{2n+3}$.

We have $a_n = \frac{1}{2n+3}$ and let $b_n = \frac{1}{n}$. Then,

$$L = \lim_{n \to \infty} \frac{a_n}{b_n} = \lim_{n \to \infty} \frac{1}{2n+3} \cdot \frac{n}{1} = \lim_{n \to \infty} \frac{1}{2 + \frac{3}{n}} = \frac{1}{2}$$

Since $0 < L < \infty$ and the harmonic series, $\sum_{n=1}^{\infty} b_n$, diverges, then so does $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{2n+3} \right|$ by the LCT.

Hence $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n+3}$ is not absolutely convergent.

Let $c_n = \frac{1}{2n+3}$. Then,

$$\lim_{n \to \infty} c_n = \lim_{n \to \infty} \frac{1}{2n+3} = 0 \quad \text{and}$$

$$c_n = \frac{1}{2n+3} > \frac{1}{2(n+1)+3} = c_{n+1}$$

Since it is decreasing, it converges by the AST.

Thus, $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n+3}$ is conditionally convergent.

CONTINUED

Theorem

<u>Ratio Test</u>

For $\sum_{n=1}^{\infty} a_n$,

(1) If $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, then it is absolutely convergent

(2) If $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$ or diverges, then it is divergent

(3) If $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, then the ratio test does not hold

Ex

Determine if the series converges or diverges.

$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ Consider $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$$= \lim_{n \to \infty} \left| \frac{(n+1)^2}{2^{n+1}} \right| \left| \frac{2^n}{n} \right|$$

$$= \lim_{n \to \infty} \frac{1}{2} \cdot \frac{\left(1 + \frac{1}{n}\right)^2}{1}$$

$$= \frac{1}{2}$$

By the ratio test, it is absolutely convergent.

Hence $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ is convergent.

ratio test

CONTINUED

Ex

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{n}$$

Consider $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$$= \lim_{n \to \infty} \left| \frac{(-2)^{n+1}}{n+1} \right| \left| \frac{n}{(-2)^n} \right|$$

$$= \lim_{n \to \infty} \left| (-2) \cdot \frac{1}{1 + \frac{1}{n}} \right|$$

$$= 2$$

By the ratio test, since $2 > 1$, it diverges.

$$\sum_{n=1}^{\infty} \frac{n^2}{n^5 + 1}$$ The ratio test fails, use LCT

$$\sum_{n=1}^{\infty} \frac{(-3)^n}{n!}$$

Consider $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$$= \lim_{n \to \infty} \left| \frac{(-3)^{n+1}}{(n+1)!} \right| \left| \frac{n!}{(-3)^n} \right|$$

$$= \lim_{n \to \infty} \left| (-3) \cdot \frac{n!}{n!\,(n+1)} \right|$$

$$= \lim_{n \to \infty} \left| \frac{(-3)}{n+1} \right|$$

$$= 0$$

Since $0 < 1$, by the ratio test, it is absolutely convergent. Hence it converges.